

Indian Institute of Technology Goa

EE 635 Digital Integrated Circuit Design

Instructor: Nandakumar Nambath

Course Project

General instructions:

- Prepare a minimal report consisting of all the results in PDF format and upload the same. You have to include the screenshots of your simulation setups in the report.
 - The report should mention the contributions of individual group members.
 - Along with the report upload a zip file containing your spice files for all the simulations.
 - Make sure that all the files you upload have your group name on them.
 - All your submissions should be less than 5 MB in size.
 - Deadlines:
 - Submission of interim report - 10 pm on 16-04-2026
 - Submission of final report - 10 pm on 23-04-2026
 - Demonstration and viva - 3 pm to 6 pm on 24-04-2026
 - You are free to assume any missing data, but state them clearly in your report.
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1. You have to implement an array multiplier that performs the multiplication of two four-bit numbers, $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$ as part of this project. The product of A and B is represented by the eight-bit number, $P = P_7P_6P_5P_4P_3P_2P_1P_0$. The circuit diagram of the array multiplier is shown in Fig. 1. You have to follow the steps given below to complete the project.
 - (a) Implement reference two-input AND, OR, and XOR gates as the basic building blocks of the multiplier. Plot the VTC and find out the propagation delays of the gates assuming that all inputs transition simultaneously.
 - (b) Instantiating the AND, OR, and XOR gates, implement a full adder. Plot the VTC and find out the propagation delays of the full adder assuming that all inputs transition simultaneously.
 - (c) Build a four-bit adder by instantiating the above full adder four times. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations.
 - (d) Implement an array multiplier by instantiating the four-bit adder and AND gates as shown in Fig. 1. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations. Calculate the power consumption and the transistor area for this implementation.
 - (e) Interim report submission: summarize the results of the above steps to a brief PDF report and submit on Moodle. (7 Marks)
 - (f) Optimize the array multiplier design such that you are able to reduce the critical path delay, or the area, or the power consumption. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations. Calculate the power consumption and the transistor area for this implementation.

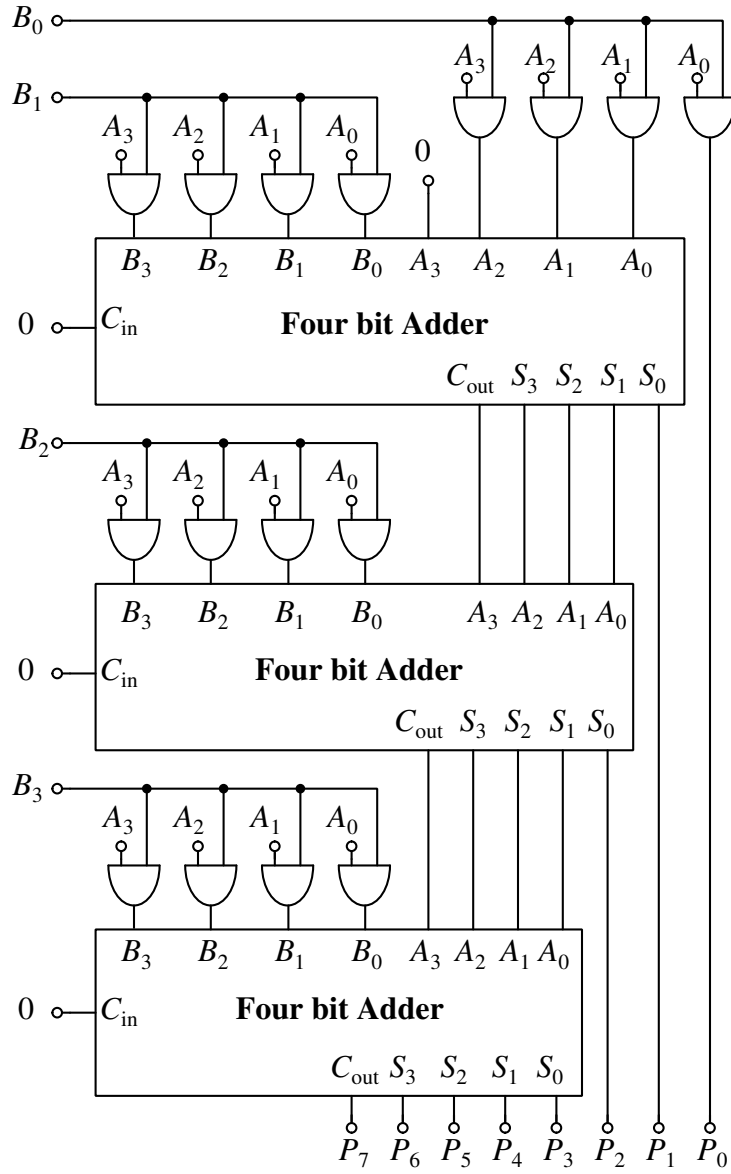


Fig. 1.

- (g) Final report submission and demonstration: summarize the results of the above step to a brief PDF report and submit on Moodle. The report should have a comparison of the results of the multiplier with and without optimization. (Marks: *Demonstration and report* - 5, *viva* - 3)